MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

III SEMESTER B.TECH REGULAR EXAMINATION NOV- 2024

SUBJECT: ENGINEERING MATHEMATICS III [MAT 2124]

Date of Exam: 27-11-2024 Time of Exam: 9.30-12.30PM Max. Marks: **50**

Q. No	Question	M
1A	Obtain a Fourier series expansion for the function f(x) given by $f(x)=\begin{cases}1+\frac{2x}{\pi}, & -\pi\le x\le 0\\ 1-\frac{2x}{\pi}, & 0\le x\le \pi\end{cases}$ and hence deduce that $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots=\frac{\pi^2}{8}$.	**3**
1B	If $f(z)=u+iv$ is analytic and given that $u-v=(x-y)(x^2+4xy+y^2)$, then determine the analytic function f(z).	**3**
1C	In what direction from the point (1,3,2) is the directional derivative of $\Phi(x,y,z)=2xz-y^2$ a maximum? What is the magnitude of this maximum?	**4**
2A	A function $f(x)$ of period 2π is specified by the following table: x: 0, $\frac{\pi}{3}$, $\frac{2\pi}{3}$, π, $\frac{4\pi}{3}$, $\frac{5\pi}{3}$, 2π f(x): 7.9, 2.2, 3.6, 0.5, 0.9, 6.8, 7.9 Obtain the Fourier Series of $f(x)$ up to the first harmonic.	**3**
2B	Evaluate $\int_c \frac{z-3}{z^2+2z+5}\,dz$, where c is the circle $(i)\ \|z\|=1$ $(ii)\ \|z+1-i\|=2$	**3**

2C	If Φ is a continuously differentiable scalar field, then show that $\nabla \times (\Phi \nabla\Phi) = 0.$	**4**
3A	Obtain the half range Fourier cosine series for the function $f(x) = \begin{cases} kx, & 0 \le x \le l/2 \\ k(l-x), & l/2 \le x \le l \end{cases}$.	**3**
3B	Show that the real valued function $u(x,y) = -\sin x \sinh y$ is harmonic.	**3**
3C	Find $F\{f(x)\}$ if $f(x) = \begin{cases} 1-\lvert x\rvert, & \lvert x\rvert < 1 \\ 0, & \lvert x\rvert > 1 \end{cases}$ and hence deduce $\int_0^\infty \left(\frac{\sin t}{t}\right)^2 \, dt = \frac{\pi}{2}.$	**4**
4A	Find the Laurent's series expansion of $f(z) = \frac{1}{(z-1)(z+3)}$ for $1 < \lvert z\rvert < 3.$	**3**
4B	Find the line integral $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region defined by $y = \sqrt{x}$ and $y = x^2$.	**3**
4C	Solve the partial differential equation $\frac{\partial^3 z}{\partial x \partial y^2} + xy + \sin(2x - 3y) = 0,$ by direct integration.	**4**
5A	Use Gauss divergence theorem to evaluate $\iint_s \vec{F} \cdot \hat{n}\, dS,$ where $\vec{F} = 4x\, i - 2y^2\, j + z^2\, k$ and S is the closed surface bounded by the region $x^2 + y^2 = 4,\ z = 0$ and $z = 3.$	**3**
5B	By the method of separation of variables, solve the wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, where c^2 is a constant.	**3**
5C	Evaluate $\oint_C \frac{e^{3iz}}{(z+\pi)^3} dz$ where C is the circle $\lvert z\rvert = 4$ in the counterclockwise direction.	**4**